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MILITARY SPECIFICATION

TEST SET, ELECTRICAL CIRCUIT, GUIDED MISSILE LAUNCHER AND AIRCRAFT, AN/ASM-149C

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the AN/ASM-149; Guided Missile Launcher and Aircraft Electrical Circuit Test Set, herein after referred to as the Test Set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation forma part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4935

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SPECIFICATIONS

Military

MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts: Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-T-21200	Test Equipment for use with Electronic and Electrical Equipment, General Specification for

STANDARDS

Military

MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-470	Maintainability Program Requirements for Systems and Equipments Prediction
MIL-STD-471	Maintainability Demonstrations
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-1385	Preclusion of Ordnance Hazards in Electromagnetic Fields; General Requirements for

HANDBOOKS

Military

MIL-HDBK-472	Military Standardization Handbook, Maintainability Predictions

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications forma part of this specification to the extent specified herein.

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BULLETINS

ANA Bulletin No. 400	Air Force-Navy Aeronautical Bulletin Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

DRAWINGS

Naval Air Systems Command

DL 1017AS100	Test Set, Electrical Circuit Guided Missile Launcher and Aircraft AN/ASM-149

MANUALS

Naval Air Systems Command

NAVAIR 17-15ASM149B-1	Maintenance Instruction (Intermediate Depot) with Calibration Instructions and IPB Missile, Launcher and Aircraft Electrical Circuit Test Set AN/ASM149B-1
NAVAIR 17-15 ASM14B-1A	Supplemental Technical Manual (Confidential)

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturer's in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Parts and materials.

3.2.1 Supplemental requirements. In the event that the contractor proposes to alter the equipment design delineated by this specification and applicable drawings, the following supplemental requirements for the selection of parts and materials are applicable.

3.2.1.1 Basic conformance. Parts and materials selected for use in the equipment shall conform to the requirements of MIL-T-21200. Parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform as required in the intended applications and environments specified herein, in which case 3.2.1 applies.

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3.2.1.2 Interchangeability. The test set shall meet the interchangeability requirements of MIL-T-21200. The drawings listed in DL 1017AS100 shall be standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.3 Design and construction. The test set shall be manufactured in accordance with DL 1017AS100, except insofar as errors of omission or commission in the data list may be incompatible with this specification. In any case, it is a condition of final acceptance that the test sets manufactured meet all the performance requirements of this specification, and appropriate test documents.

3.3.1 Reliability.

3.3.1.1 Operational stability. The test set shall operate with required performance, continuously or Intermittently for a period of at least 50 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use (see 4.4.1.2).

3.3.1.2 Operating life. The test set shall have a total operating life of 10,000 hours with minimum servicing and replacement of parts.

3.3.1.3 Reliability in mean-time-between-failures (MTBF). The test set shall have a minimum of 300 hours of mean (operating) time between failures when tested and accepted as specified in 4.5.3.

3.3.2 Radio interference. The generation of radio interference by the equipment and vulnerability of the test set to radio interference shall be within the limits of MIL-I-6181 except that broadband switching transient interference may exceed the limits as follows:

a. Conducted 50 db above 1 microvolt/mhz

b. Radiated below 25 mhz 30 db above 1 microvolt/mhz

c. Radiated above 15 mhz 10 db above 1 microvolt/mhz

The test set shall operate within the specified tolerance limits specified in 3.5 while being operated in a radio frequency interference environment as specified in MIL-STD-1385.(see 4.6)

3.3.3 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

a.	Temperature	Room ambient (30 ± 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient
e.	Input power voltage	115 ± 1.5 volts, 400 ± 5 Hz ac

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3.3.4 Service conditions. The test set shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-T-21200 for class 2 equipment, except as modified herein.

3.3.4.1 Shock. In addition to the shock requirements of MIL-T-21200, the test set when stored in its combination case, shall operate as specified in 3.5 after subjection to two 1/2 sine wave, 70g shocks, each of seven milliseconds duration applied upward along the vertical axis of the test equipment case. Physical damage to the equipment is permissible provided the operational requirements specified in 3.5 are maintained.

3.3.5 Warm-up time. The test set shall require no warm-up time under any of the service conditions specified in 3.3.4.

3.3.6 Primary input power. The test set shall be capable of specified operation when supplied input power having characteristics specified in MIL-T-21200. The power required shall be of the following type and shall be not greater than the specified amount.

AC Power, Single Phase, 115 volts, 380-420 Hz, 95 volt-amperes

3.3.7 Mean-time-to-repair (MTTR). The mean-time-to-repair (MTTR) of the equpiment shall be not greater than 1.5 hours when replacing circuit modules, plug-ins, assemblies and parts that are intended for replacement at the intermediate level.

3.4 Performance. The performance of the test set shall be as specified below plus any and all performance requirements of DL 1017AS100. Unless otherwise specified herein, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variations will be specified by the goverment.

3.5 Detail requirements.

3.5.1 Test Set, Electrical Circuit, Guided Missile Launcher and Aircraft AN/ASM-149A. The test set shall be manufactured in accordance with DL 1017AS100 and shall meet the requirements stated thereon and in the following subparagraphs.

a. All applied voltages, loads and measurements shall be referenced to J101-A (ground) unless otherwise specified herein.

b. When a requirement specifies that a voltage is applied to J101B, circuit breaker CB101 must be ON to obtain the required indication.

3.5.1.1 Electrical requirements. The test set shall meet the following electrical requirements:

3.5.1.1.1 Panel illumination. With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, the illuminating panel shall be illuminated in all positions of the TEST switch.

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3.5.1.1.2 Interlock circuit. With 115 ± 11.5 volts AC, 400 ± 20 Hz applied through a 2600 ± 260-ohm resistance to J101-R, the maximum current into J101-R for TEST switch positions 4 and 16 (OFF) shall be 1 ma. For all other TEST switch positions, the minimum current into J101-R shall be 35 ma.

3.5.1.1.2.3 Headtone output. With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, the headtone output shall be present at J101-U for TEST switch positions 5 through 10 unless otherwise specified herein. The headtone output shall be a pulse whose repetition rate has a period corresponding to one of the four states listed below, where the nominal, minimum, and maximum values of T1, T4, T2 and T3, and T3 are specified in NAVAIR 17-15ASM149B-1A (CONFIDENTIAL). For HEADTONE FUNCTION switch positions CYC SIDS and CYC SLOW the headtone output shall be cycling through the four states in a ...-3-4-1-2-3-4-1-2... sequence with a dwell time of 10 ± 2 seconds in each state. Placing the HEADTONE FUNCTION switch in position NON CYC SIDS or NON CYC SLOW shall stop the cycling and hold the existing state. Cycling shall resume at the point of interruption when the HEADTONE FUNCTION switch is returned to position CYC SIDS or CYC SLOW. The headtone output pulse shall have the following characteristics with a resistive load of 220 ± 22 ohms applied to J101-U:

TABLE I. Headtone output pulse characteristics.

Period:	
State 1	T1
State 2	T4
State 3	T2 and T3
State 4	T3
Pulse Width (at 50 percent of peak amplitude)	180 ± 10 microseconds
Pulse Amplitude (peak)	+6.6 ± 1.2 volts
Pulse Rise Time	5 microseconds max.

3.5.1.1.4 TEST switch position 1 (heater voltage). An input voltage of 104.0 ± 1.0 volts AC, 400 ± 20 Hz applied to J101-B shall cause the QUALITY meter to indicate on the center of the line which determines the left edge of the triangle zone. When the input voltage at J101-B is increased to 126.5 ± 2.0 volts AC, the QUALITY meter shall indicate on the center of the line which determines the right edge of the triangle zone. The current into J101-B shall be 0.60 ± 0.20 amperes for any input voltage from 103.5 to 126.5 volts AC.

3.5.1.1.5 TEST switch position 2 (guidance and control power on). An input voltage of 104.0 ± 1.0 volts AC, 400 ± 20 Hz applied to J101-C shall cause the QUALITY meter to indicate on the center of the line which determines the left edge of the triangle zone. When the input voltage at J101-C is increased to 126.5 ± 2.0 volts AC, the QUALITY meter shall indicate on the center of the line which determines the right edge of the triangle zone. The input resistance at J101-C shall be 230 ± 15 ohms.

3.5.1.1.6 TEST switch position 3 guidance and control power off). An input voltage of 104.0 ± 1.0 volts AC, 400 ± 20 Hz applied to J101-C shall cause the QUALITY meter to indicate on the center of the line which determines the left edge of the triangle zone. When the input voltage at J101-C is

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increased to 126.5 ± 2.0 volts AC, the QUALITY meter shall indicate on the center of the line which determines the right edge of the triangle zone. The input resistance at J101-C shall be 230 ± 15 ohms.

3.5.1.1.7 TEST switch position 4 (heater voltage interlock). An input voltage of 104.0 ± 1.0 volts AC, 400 ± 20 Hz applied to J101-B shall cause the QUALITY meter to indicate on the center of the line which determines the left edge of the triangle zone. When the input voltage at J101-B is increased to 126.5 ± 2.0 volts AC, the QUALITY meter shall indicate on the center of the line which determines the right edge of the triangle zone. The current into J101-B shall be 0.60 ± 0.20 amperes for any input voltage from 103.5 to 126.5 volts AC.

3.5.1.1.8 TEST switch position 5 (target direction indicator foresight). With 115 ± 11.5 volts Ac, 400 ± 20 Hz applied to J101-B, the QUALITY meter shall indicate in the circle zone. The headtone output shall be present as specified in 3.5.1.2.3 and the direction finder outputs shall be in accordance with table II.

3.5.1.1.9 TEST switch position 6 (target direction indicator left). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, the QUALITY meter shall indicate in the circle zone. The headtone output shall be present as specified in 3.5.1.2.3 and the direction finder outputs shall be in accordance with table II.

3.5.1.1.10 TEST switch position 7 (target direction indication down). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, the QUALITY meter shall indicate in the circle zone. The headtone output shall be present as specified in 3.5.1.2.3 and the direction finder outputs shall be in accordance with table II.

3.5.1.1.11 TEST switch position 8 (fire sequence). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, the headtone output and the QUALITY meter indication shall be in accordance with table III with 28 ± 2 volts DC applied to J101-Z, the input resistance at J101-H shall be 25 ± 2.5 ohms.

TABLE III. Headtone output and QUALITY meter indication.

Input		Output	
J101-H	J101-Z	Headtone per 3.5.1.2.3	QUALITY meter indication
0	0	Yes	None
0	28 ± 2 volts DC	No	None
22.5 ± 0.5 volts DC	28 ± 2 volts DC	No	Left edge of Circle Zone
31.0 ± 1.2 volts DC	28 ± 2 volts DC	No	Right edge of Circle Zone
28 ± 2 volts DC	0	Yes	None

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TABLE II. Direction finder outputs.

TEST Switch Position	HEADTONE FUNCTION Switch Position	Pin 1/			
		J101-V	J101-X	J101-W	J101-Y
5	CYC SLOW or NON CYC SLOW	5.00 0.20 volts DC	5.00 0.20 volts DC	5.00 0.20 volts DC	5.00 0.20 volts DC
		Difference 0	0.01 volt DC	Difference 0	0.01 volt DC
	CYC SIDS or NON CYC SIDS	5.00 0.20 volts DC with 0.62 0.20 volt p-p	5.00 0.20 volts DC with 0.62 0.20 volt p-p	5.00 0.20 volts DC with 0.62 0.20 volt p-p	5.00 0.20 volts DC with 0.62 0.20 volt p-p
		Difference 0	0.01 volt DC, with 1.25 0.25 volts p-p	Difference 0	0.01 volt DC, with 1.25 0.25 volts p-p
6	CYC SLOW or NON CYC SLOW	5.00 0.20 volts DC	5.00 0.20 volts DC	1.00 0.20 volt DC	9.00 0.20 volts DC
		Difference 0	0.01 volt DC	Difference 8.00	0.10 volts DC
	CYC SIDS or NON CYC SIDS	5.00 0.20 volts DC with 0.62 0.20 volt p-p	5.00 0.20 volts DC with 0.62 0.20 volt p-p	1.00 0.20 volt DC with 0.62 0.20 volt p-p	9.00 0.20 volts DC with 0.62 0.20 volt p-p
		Difference 0	0.1 volt DC, with 1.25 0.25 volts p-p	Difference 8.00	0.10 volts DC, with 1.25 0.25 volts p-p

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TABLE II. Direction finder outputs - continued

TEST Switch Position	HEADTONE FUNCTION Switch Position	Pin 1/			
		J101-V	J101-X	J101-W	J101-Y
7	CYC SLOW or NON CYC SLOW	1.00 0.20 volt DC	9.00 0.20 volts DC	5.00 0.20 volts DC	5.00 0.20 volts DC
		Difference 8.00 0.10 volts DC		Difference 0 0.01 volt DC	
	CYC SIDS or NON CYC SIDS	1.00 0.20 volt DC with 0.62 0.20 volt p-p	9.00 0.20 volts DC with 0.62 0.20 volt p-p	5.00 0.20 volts DC with 0.62 0.20 volt p-p	5.00 0.20 volts DC with 0.62 0.20 volt p-p
		Difference 8.00 0.10 volts DC, with 1.25 0.25 volt p-p		Difference 0 0.01 volt DC, with 1.25 0.25 volt p-p	

1/ Outputs shall be as specified when a load resistance of 20 K 1 K ohms is applied from J101-V to J101-X and a load resistance of 20 K 1 K ohms is applied from J101-W to J101-Y. The AC voltage shall be a sine wave with period T5 as specified in NAVAIR 17-15ASM149B-1A (CONFIDENTIAL).

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3.5.1.1.12 TEST switch position 9 (firing ready). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied at J101-B and 28 ± 2 volts DC applied to J101-Z, the headtone output shall not be present at J101-U. An input voltage of 22.5 ± ().5 volts DC applied to J101-H shall cause the QUALITY meter to indicate on the center of the line which determines the left edge of the circle zone. When the input voltage at J101-H is increased to 31.0 ± 1.2 volts DC, the QUALITY meter shall indicate on the center of the line which determines the right edge of the circle zone. The input resistance at J101-H shall be 25 ± 2.5 ohms. When the voltage is removed from J101-Z, the headtone output shall be present at J101-U.

3.5.1.1.13 TEST switch position 10 (firing signal). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B and 28 ± 2 volts DC applied to J101-Z, the headtone output shall not be present at J101-U. An input voltage of 22.5 ± 0.5 volts DC applied to J101-Z shall cause the QUALITY meter to indicate on the center of the line which determines the left edge of the circle zone. When the input voltage at J101-Z is increased to 31.0 ± 1.2 volts DC, the QUALITY meter shall indicate on the center of the line which determines the right edge of the circle zone. The input resistance at J101-Z shall be 23.8 ± 2.5 ohms. When voltage is removed from J101-Z, the headtone shall be present at J101-U.

3.5.1.1.14 TEST switch position 11 firing ready stray voltage). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, an input of 0.065 ± 0.005 volts AC or ± 0.065, ± 0.005 volts DC applied to J101-H shall cause the QUALITY meter to indicate in the rectangle zone. When the input voltage at J101-H is increased to 0.100 ± 0.005 volts AC or ± 0.100, ± 0.005 volts DC, the QUALITY meter shall indicate in the BAD zone between the rectangle and circle zones.

3.5.1.1.15 TEST switch position 12 (firing signal stray voltage). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, an input of 0.065 ± 0.005 volts AC or ± 0.065, ± 0.005 volts DC applied to J101-Z shall cause the QUALITY meter to indicate in the rectangle zone. When the input voltage at J101-Z is increased to 0.100 ± 0.005 volts AC or ± 0.100, ± 0.005 volts DC, the QUALITY meter shall indicate in the BAD zone between the rectangle and circle zones.

3.5.1.1.16 TEST switch position 13 (meter test). With 115 ± 11.5 volts AC, 400 ± 20 Hz applied to J101-B, the QUALITY meter shall indicate in the circle zone.

3.5.1.1.17 TEST switch position 14 (master arm). An input voltage of 22.5 ± 0.5 volts DC applied to J101-G shall cause the QUALITY meter to indicate on the center of the line which determines the left edge of the circle zone. When the input voltage at J101-G is increased to 31.0 ± 1.2 volts DC, the QUALITY meter shall indicate on the center of the line which determines the right edge of the circle zone. The input resistance at J101-G shall be 25 ± 2.5 ohms.

3.5.2 Simulator, Missile Firing SM-465/ASM-149A. The simulator shall be manufactured in accordance with the applicable drawings of DL 1017AS100. The simulator shall be capable of providing a means of checking the aircraft intervalometer.

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3.5.3 Cable Assembly, Special Purpose, Electrical CX-9297/ASM-149. The cable assembly shall be manufactured in accordance with the applicable drawings of DL 1017AS100. The cable assembly shall be capable of connecting the front panel connector J101 of the test set to the SHRIKE connector of the launcher under test or to Adapter, Connector U-301/ASM-149 when the launcher is not in place.

3.5.4 Adapter, Connector U-301/ASM-149. The adapter shall be manufactured in accordance with the applicable drawings of DL 1017AS100. The adapter shall be capable of providing a means of adapting connector P104 of Cable Assembly CX-9297/ASM-149 to the aircraft wing connector when the launcher is not in place.

3.5.5 Case, Test Set CY-6306/ASM-149A. The case shall be manufactured in accordance with the applicable drawings of DL 1017AS100. The case, including a cover, shall be capable of being used to the house the test set, the simulator, the cable assembly, the connector adapter, the hexagon wrench and the handbook of maintenance and calibration instructions.

3.5.6 Wrench, hexagon key, 58A5A294-4. The wrench shall be manufactured in accordance with the applicable drawings of DL 1017AS100. The wrench shall be capable of being used to remove and reinstall the set screws in the knob of the TEST switch on the front panel of the test set during maintenance operations.

3.6 Soldering. Soldering shall be in accordance with MIL-STD-454, requirement 5.

3.7 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified herein all inspections shall be performed in accordance with the test conditions specified in 3.3.3 and 3.3.4.

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TABLE IV. Inspection requirements

Requirements	I. First Article Inspection	II. Quality Conformance Inspection			
		Individual	Sampling	Special	Life
Visual Examination (3.2 through 3.3)	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1
Performance Characteristics (3.4, 3.5)	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2
Workmanship (3.7)	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3 See 4.5.6.2 and 4.5.6.3
Environmental Tests (3.3.4)	4.7				
Warm-up time (3.3.5	4.8	1/	2/	3/	
Interference Control (3.3.2)	4.6				
Reliability (3.3.1)	4.5.3		4.5.3		
Maintainability (3.3.7)	4.9		4.9		

1/ Additional individual tests may be requested by the procuring activity (see 4.5.1).

2/ Sampling tests to be performed shall be determined by the procuring activity based on first article and individual test results (see 4.5.2.1).

3/ Special tests to be performed shall be determined by the procuring activity (see 4.5.4.2).

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4.4 First article inspection. First article inspection shall be conducted by the contractor on one test set representative of the production equipments to be supplied under the contract. First article tests shall be accomplished under the approved test procedures (see 4.7). Column I of table IV lists minimum tests required for first article inspection. The procuring activity shall be advised through the goverment inspector when tests are to be conducted so that a government representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government. No first article inspections shall be conducted prior to approval of the first article test procedure by the contracting officer (see 6.2.2).

4.4.1 First article test data. The contractor shall make available data collected in conducting first article tests for procuring activity review (see 6.2.2). On first production contracts this data shall include a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures. The ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

4.4.2 Scope of tests. First article tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of this specification. First article tests shall include environmental tests in accordance with the requirements of MIL-T-5422 to the limits specified in MIL-T-21200 for class 2 equipment except as modified by 3.3.4.1. In addition, first article tests shall include interference tests in accordance with MIL-I-6181, tests to assure operational capability in environments specified by MIL-STD-1385, and drop and water-tightness tests in accordance with MIL-T-21200.

4.4.3 First article approval. Approval of the first article sample shall be by the procuring activity upon completion of all tests and the first article test report being made available (see 6.2.2). No production equipments shall be delivered prior to the approval of the first article sample. The first article sample shall not be considered as one of the equipments under the contract.

4.4.4 Production equipments. Test sets supplied under the contract shall in all respects, including design, construction, workmanship, performance, and quality, be equal to the approved first article sample. Each equipment shall be capable of successfully passing the same tests as imposed on the first article sample. Evidence of non-compliance with the above shall constitute cause for rejection of a test set and of test sets already accepted by the Government. It shall be the obligation of the contractor to make necessary corrections as authorized by the procuring activity.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection. All inspection and testing shall be under the supervision of the government inspector. Contractors not having adequate facilities for conducting all required tests shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall prepare and make available test reports showing quantitative results for all quality conformance tests (see 6.2.2). Such reports shall be signed by an authorized

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representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

a. Individual Tests (see 4.5.1)

b. Sampling Tests (see 4.5.2)

c. Reliability Assurance Tests (see 4.5.3)

d. Special Tests (see 4.5.4)

e. Life Test (see 4.5.6)

4.5.1 Individual tests. Each test set submitted for quality conformance inspection shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship. operational adequacy and reliability. As a minimum, each test set accepted shall have passed the following tests:

a. Examination of Product (see 4.5.1.1)

b. Operational Test (see 4.5.1.2)

c. Manufacturing Run-In Test (see 4.5.1.3)

4.5.1.1 Examination of product. Each test set shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each test set shall be operated long enough to permit the test set temperature to stabilize and to check sufficient characteristics and record adequate data to assure specified operation of the test set.

4.5.1.3 Manufacturing run-in test. Each test set shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the test set. Only those adjustments will be permitted which can be made by using controls and adjustments that are accessible to the operator during the normal use of the test set.

a.	Temperature	Ambient room
b.	Humidity	Ambient room
c.	Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

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The test set shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the six period of operation. Where feasible, the test set shall be operated during this vibration for the purpose of detecting flaws and imperfect workmanship. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the test set in two directions the vertical direction shall be used. During the six hour period of operation following the 10-minute vibration period, the test set shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future test set. A record shall be kept of all failures. The six hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

4.5.2 Sampling tests. Test sets selected for sampling tests shall first have passed the individual tests. Test sets will be selected for sampling tests by the government inspector in accordance with table V.

4.5.2.1 Scope of sampling tests. As a minimum, each test set selected for sampling tests shall be subjected to the following tests:

TABLE V. Sampling tests sample selection.

Quantity of Test Sets Offered for Acceptance	Quantity to be Selected for Sampling Test 1/
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

1/ Sampling Tests are not required when Reliability Assurance Tests are conducted.

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each test set tested and should be based on results of the first article, individual, and special tests.

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c. Manufacturing run-in test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future test sets.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the Qualification Phase and the Production Acceptance (Sampling) Phase shall be conducted.

4.5.3.1 Qualification phase. Prior to the acceptance of test sets under the contract or purchase order, a minimum of three test sets shall be tested in accordance with MIL-STD-781, under the section entitled “Qualification Phase of Production Reliability Tests”. The maximum number of test sets to be used shall be those listed in table V of MIL-STD-781. For the qualification phase, Test Level E shall be used. The accept-reject criteria for Test Plan IV shall be used.

4.5.3.2 Reliability sampling phase tests. Samples of the test set shall be tested in accordance with MIL-STD-781 under the section entitled “Sampling Phase of Production Reliability Tests”. For the sampling phase, Test Level E shall be used. The accept-reject criteria for Test Plan VI shall be used to determine the length of the tests until an accept or reject decision is reached.

4.5.3.2.1 Lot size for sampling phase. The entire contract quantity (with the exception of those test sets used in the Qualification Phase) shall be considered one lot. The test program shall start the first month after the Qualification Phase has been completed. Samples shall be selected and placed on test in the same manner as though one month’s production was a lot. The samples shall be tested until an accept or reject decision is reached. In the meantime other test sets constructed shall be shipped. The test results shall be summarized monthly for the procuring activity (see 6.2.2). The procuring activity reserves the right to stop the acceptance of test sets any time after one or more reject decisions have been reached, pending a review of the contractor’s efforts to improve the test sets, the test sets parts, the test set quality control, etc., so that the entire contract quantity will show an accept decision.

4.5.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be made available and accepted by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests (see 6.2.2).

4.5.4 Special tests. Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the test set and to assure adequate quality control. The test sets selected for special tests may be selected from test sets previously subjected to the sampling or reliability assurance tests.

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4.5.4.1 Special test schedule. Selection of test sets for special tests shall be amde as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.5.4.2 Scope of special tests. Special tests shall consist of such tests as are acceptable to by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and make available for procuring activity acceptance prior to conducting the tests (see 6.2.2).

4.5.5 Test Set. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or a design defect.

c. Have available for procuring activity authorization, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests (see 6.2.2).

d. Where practical, include a test in the individual test to check all test sets for this requirement until assurance is obtained that the defect has been corrected.

4.5.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on test sets that have passed the individual tests. The life test shall be performed under the conditions specified in 4.5.6.1. The life test sample will be selected by the government inspector in accordance with the following. (Test Sets which have successfully passed the Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of test sets equal to, or more than, that listed in table VI receive at least 300 hours each of test time.

4.5.6.1 Test conditions. The life test shall be conducted under the standard conditions specified in 3.3.3.

4.5.6.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the test set temperature to stabilize. Periodically, the test set shall be turned on and off several times and put through its various phases of operation.

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TABLE VI. Life tests sample selection.

Quantity of Test Sets Offered for Acceptance	Quantity to be Selected for Life Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.5.6.3 Performance check. At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check shall be subject to acceptance by the procuring activity (see 6.2.2).

4.5.7 Test data. The contractor shall keep a daily record of the performance of the test sets, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test. This record shall indicate the following:

a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.5.7.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be prepared and made available for the procuring activity upon completion of the test (see 6.2.2). In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.6 Electromagnetic interference (EMI) tests. EMI tests shall be performed in accordance with MIL-I-6161 to determine that the test set meets the requirements specified in 3.3.2.

4.7 Environmental tests. The test set shall be subjected to environmental tests in accordance with MIL-T-21200 for class 2 equipment.

4.8 Warm-up time. Warm-up time shall not be required prior to recording performance test data.

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4.9 Maintainability test. The Mean-Corrective-Maintenance-Time and Maximum-Corrective-Maintenance as specified in 3.3.7 shall be demonstrated as specified in MIL-STD-471, Test Method 9.

4.10 Test procedures. The procedures used for conducting first article tests and quality conformance tests shall be prepared and made available by the contractor for procuring activity review and acceptance. (see 6.2.2). The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.2). MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures will be provided and may be used when their use is authorized by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary (see 6.2.1).

4.11 Presubmission testing. No item, part or complete test set shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

5. PACKAGING

5.1 General. All major units and parts of the test sets shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or purchase order in accordance with MIL-E-17555 (see 6.2.1).

6. NOTES

6.1 Intended use. The AN/ASM-149C Guided Missile Launcher and Aircraft Electrical Circuit Test Set AN/ASM-149C is intended for use to determine the operational capability and safety of aircraft and launcher circuitry employed in the launching of the AGM-45A (SHRIKE) missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1).

c. Whether first article inspection is required (see 3.1 and 6.3).

d. If additional individual tests are required (see 4.5.1).

e. Lot acceptance for each lot, if other than specified in 4.5.3.2.1.

f. Test procedure modification or if additional tests are required (see 4.10).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporate a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (N) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph no.	Data requirement title	Applicable DID no.	Option
4.4, 4.10	Procedure, First Article Inspection	DI-T-4901	--
4.4.1, 4.4.3	Report, First Article Inspection	DI-T-4902	--
4.5.4.2, 4.5.5c, 4.5 4.7	Procedures, Production/Acceptance Inspection	DI-T-4903	--
4.5.3.3	Procedures, Reliability Tests	DI-R-7035	--
4.5.7.1	Report, Failure	UDI-R-21141A	--

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required the item will be tested and should be a first article sample. The first article should be one test set. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination and approval of the documents' first article.

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-18404B	Test Set, Electrical Circuit, Guided MIssile Launcher and A/C

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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